

THIAGO MEMELLI

AI Engineer | Python | LLM Systems | Backend Engineering

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PROFESSIONAL SUMMARY

Software engineer with more than 13 years of experience developing, supporting, and evolving enterprise systems, including mission-critical and 24/7 operating environments.

Currently focused on AI Engineering, building applications and infrastructure for LLM-based systems with Python, FastAPI, asynchronous processing, and integration with multiple AI providers.

Recent hands-on experience building resilient inference APIs, asynchronous concurrency, retries, model and provider fallbacks, rate limiting, failure handling, typed contracts, automated testing, and deployment of real applications.

Special interest in LLM Engineering, RAG, embeddings, vector search, and building production-oriented AI systems.

FEATURED PROJECT

LLM Inference Service

AI Engineering / LLM Infrastructure

Demo: app.thiagomemelli.com.br | **API:** api.thiagomemelli.com.br/docs

Asynchronous inference service built with Python and FastAPI to abstract and make access to multiple LLM providers resilient.

- Multi-provider integration with Groq and Gemini
- Per-provider concurrency control
- Exponential backoff with jitter
- Model fallback
- Fallback cycle protection
- Per-client rate limiting
- Token, provider, model, and latency metrics
- Public API documented with OpenAPI / Swagger
- 44 automated tests
- Asynchronous processing with Asyncio
- Retry for transient failures
- Per-provider timeout
- Provider fallback
- SDK error classification and normalization
- Request ID tracing
- Data and configuration validation with Pydantic
- Public frontend integrated with the API
- Static analysis and code quality with mypy and Ruff

Stack: Python • FastAPI • Asyncio • Pydantic • Groq API • Gemini API • Pytest • mypy • Ruff • REST APIs • OpenAPI

OTHER PROJECTS

CashFlow API

Financial API built with Python and FastAPI for managing income, expenses, categories, and reports. Includes an AI assistant integrated with OpenAI to answer questions using the user's financial data.

- REST API with FastAPI
- OpenAI API integration
- JWT authentication
- SQLAlchemy ORM
- Pydantic validation
- Docker
- 67 automated tests with Pytest
- OpenAPI / Swagger documentation

Stack: Python • FastAPI • OpenAI API • SQLAlchemy • Pydantic • Pytest • Docker • JWT

Sales Insights Pipeline

Automated sales data processing and analytics pipeline built with Python. Performs CSV validation and cleaning, financial metric calculation, outlier detection, automatic insight generation, and report and visualization creation. Documented benchmark of approximately 8.5 seconds for processing 100,000 records.

Stack: Python • Pandas • NumPy • Matplotlib • Seaborn • OOP • Data Processing

TECHNICAL SKILLS

AI Engineering: LLM APIs • Multi-provider LLM Systems • LLM Inference • Prompt Engineering • Retry & Fallback Strategies • Error Handling • Rate Limiting

Python & Backend: Python • FastAPI • Asyncio • Pydantic • SQLAlchemy • REST APIs • OpenAPI • JWT

Architecture & Quality: OOP • Type Hints • Protocols • Dataclasses • Pytest • mypy • Ruff • Async Concurrency

Data: SQL • SQLite • SQL Server • Data Modeling

Tools & Environment: Linux • Git • GitHub • Docker • VS Code

Currently deepening: RAG • Embeddings • Vector Search • Vector Databases • Redis

PROFESSIONAL EXPERIENCE

AI Engineering / Python Backend - Independent Projects

August 2025 - Present

Developing projects focused on modernizing my technical stack and specializing in AI Engineering.

- Development and deployment of a multi-provider LLM inference service
- Building asynchronous APIs with Python and FastAPI
- Implementation of retry, fallback, timeout, concurrency, and rate-limiting mechanisms
- Integration with language model APIs
- Development based on typed contracts and separation of responsibilities
- Creation of automated tests and static code validation
- Development and deployment of internet-accessible applications
- Building hands-on projects focused on Backend, AI, and data processing

International Experience & Hospitality

June 2018 - July 2025

Relocated to Europe and worked in the hospitality sector in restaurant service and kitchen roles.

Worked as both waiter and cook, developing strong customer service, teamwork, adaptability, communication in multicultural environments and the ability to perform under pressure in fast-paced operations.

In 2025, I returned to software engineering with a focused transition into AI Engineering, Generative AI and modern Python backend systems.

Senior Systems Analyst

SPG Negócios de TI e Telecom | May 2006 - June 2018 | Vitória, ES

Worked on highly critical 24/7 systems for major telecommunications companies, including Oi and Brasil Telecom.

- Support and evolution of critical enterprise systems
- Analysis and resolution of production incidents
- Systems development and maintenance
- Integration across platforms and systems
- Investigation of performance issues
- Work with databases and large-volume data processing
- Participation in technical analysis and solution design

Environment: Natural/Adabas • SQL Server • SOA • Enterprise Systems

Systems Developer

InfoSis Consultoria | March 2005 - March 2006 | Vitória, ES

- Application development
- Systems analysis
- Development with Delphi and SQL Server
- Data modeling
- Requirements gathering with clients

EDUCATION

MBA in Project Management - Fundação Getulio Vargas (FGV), 2009 - 2011

Bachelor's Degree in Computer Science - FAESA, 2002 - 2005

LANGUAGES

Portuguese: Native

English: Intermediate - Good technical documentation reading proficiency and experience working with tools, APIs, and documentation in English.

AREAS OF INTEREST

AI Engineering • LLM Engineering • RAG Systems • Generative AI • Python Backend • AI Infrastructure

Work arrangement: Remote • On-site • Hybrid